The Onduscular Equations

For the electron equation, we start from the Onduscular theory ("onde" and "corpuscle" in French) is refers to an elementary particle with no mass and which is moving at light speed and acts like a wave and particle: photon, graviton, monopole, particle$_T$, etc…and particle that has mass and a speed smaller than light (atoms nucleus, positron, electron, particle$_M$…). The onduscular equation $\frac{2m}{h^2}(E-U(r))f(r,t)+\frac{R}{r}\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (1) is valid in both spherical and cylindrical coordinates it emerges from a combination of Schrodinger which is the first degree atemporal and Klein-Gordon temporal equation of the second degree with parameters R chosen so for a small radius $r_{Bohr,}$ for atomic value to coincide with the Schrodinger equation exactly and predict the evolution of wave function to huge values of r about the galactical distances. The wave function of the Schrodinger equation has the supposition that is null at infinity, is normed, has the dimension of [$L^{-3/2}$], and predicts the probability of the existence of the microparticle in volume dV. Schrodinger eq is: $\frac{2m}{h^2}(E-U(r))f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=ih\frac{\partial f(r,t)}{\partial t}$ (2) & Klein-Gordon equation is $\frac{2m}{h^2}f(r,t)+\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (3) thus eq (1) for m = 0 is a *wave equation* with *variable parameters* and seems to be true $\frac{R}{r}\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (4). But all 3 equations are different and also wave equation is not satisfactory $\frac{\partial^2(f(r,t))}{\partial r^2}=\frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2}$ (5),

The onduscular wave function of onduscular equation has the supposition as null at R (or 2R) is normed, has the dimension of [$L^{-3/2}$], and predicts the probability of the existence of the microparticle in volume dV. Also, a condition of the derivate = R in origin verifies the condition that the speed of interaction has the velocity V whatever will be that. The value of R=3.9228·10^{22} m is calculated in OndSLP.pdf: The change of variable from Schrodinger to QFT Document

As you can see function $f_1(r,t) = rf(r,t) = rg(r)u(t)$ is the Onduscular Wave Function (OWF) multiplied by r denoted f(r,t) in the case of graviton equation and other particles that travel with the speed of light so we call it Effective Onduscular Wave Function or EOWF. The EOWF has the dimension of [$L^{-1/2}$], and OWF has the dimension of [$L^{-3/2}$] with separate variables and g(r) has dim of [$L^{-3/2}$], u(t) is adimensional as in the Schrodinger equation. Both equations Onduscular and Schrodinger have almost the same value of the function that depends on r, g(r) multiplied by an adimensional exponential factor that depends on time t. The relation between f1(r,t) EOWF and f(r,t) OWF is $f_1(r,t) = rf(r,t) = rg(r)u(t)$ thus we calculus EOWF which is easier, and we return to the OWF by dividing by r the function $f_1(r,t) \Rightarrow f(r,t) = f_1(r,t)/r$.

The duality wave-particle is generated by the fact that all atoms and particles receive energy from stars and are one-time quanta particles and one-time quanta waves of probability and have the definition $p(x,t)=|\psi(x,t)|^2$ as in Bohm's Quantum Potential and between states executes an instant quantum jump from the successive position in space due to the granularity of time. Also suppose that force-mediating particles of these interactions are electronic neutrinos for gravity which is a Majorana particle with spin quantum number s = 0, photons for light, monopoles for magnetic forces, and tau neutrino for time which is a Dirac particle with spin quantum number s = -½ ... Space has a special treatment but we bearly say that muon neutrino is the place-mediating particle which is a Dirac particle with spin quantum

number s = -½. The other elementary stable particles are the electron, proton, neutron, and phonon which are treated as an equation in TOnduscular.pdf;
Schrodinger's and onduscular equations inference document
The initial derivate = R, the boundary condition f(R)=0 or f(2R)=0, or another initial condition was determined by dowsing. The suppositions that I make are in the file Supposition.pdf.

https://michaelvio.github.io/Supposition.pdf
The Onduscular Wave Function (OWF) is the value of a particle into space-time at instant t into spherical, cylindrical, or Cartesian cords specific for every particle in particular: for the graviton is for a spatial wave function $[L^{-3/2}]$ and of the function of graviton f (r,t) is $[L^{-3/2}]$ ...and is related to the probability to find the graviton in the local space from 0 to 2·R (where R = $3.9228 \cdot 10^{22}$ m) these are onduscular wave function concrete value ("valori concrete ale function de unda onduscular" in Romanian language) as you can see the f is bounded function ("marginita") and we can normalize that function. Besides these values 0 to 2·R for the graviton the wave function has improper virtual values ("Valori improprii virtuale") and the function is not bordered in fact for r over 2R the force of attraction change sign, goes to infinity when r->-∞ (repulsion) and extremely high but not ∞ in origin. At the normalization condition that is integral with respect to r and for t fix, t = t_0 we have: $4\pi \int_0^{2R} \psi(r,t_0)^2 dr = 1$ where ψ(r,t)=f(r,t)/r.
In the graviton equation f(r,t)= u(t)·g(r); where $[L]^{-3/2}$ and the probability to find the graviton into the volume dV is P(r,t) = $|\psi(r,t)^2|dV$ in 3-dimensional space as in Bohm's Quantum Potential. To and from the condition that the velocity of all the below particles, except the electron, is c we recalculate the basis-separated solution and the singular solution of onduscular equation each satisfying a set of three singular conditions. We mention that the below denoted k_1, k_2, k_3, k_4, k_5 are real constants, which are specific to the particle. We supposed that all the functions are separable into a produced single function by one variable. The duality wave-particle is generated by the fact that all atoms and particles receive energy from stars and are one-time quanta particles and one-time quanta waves of probability and have the definition $p(x,t) = |\psi(x,t)|^2$ as in Bohm's Quantum Potential $P(r_b,t_0) = |\psi(r,t_0)|^2$ and between states executes an instant quantum jump from the successive position in space due to the granularity of space-time. Also, suppose that force-mediating particles of these interactions are electronic neutrinos for gravity which is a Majorana particle with spin quantum number s = 0, photons for light s = 1, monopoles for magnetic forces, and tau neutrino for time, a Dirac microparticle with spin quantum number s = -½ ... Space has a special treatment but we barely say that muon neutrino is the space-mediating particle, a Dirac particle with spin quantum number s = -½. The other elementary stable particle is the electron, proton, neutron, and phonon which are treated as an equation in TOnduscular.pdf; As in Bohm's pilot-wave theory we have the Onduscular equation instead of Schrodinger and flux conservation of first or second degree homogeneous or nonhomogeneous instead of guiding equation $v = \frac{Jx}{p} = \frac{\hbar}{m} Im[\frac{1}{\Psi}(\frac{\partial \Psi}{\partial x})]$
The theory of everything should describe all the characteristics of the moving particle into the space coordinate and temporal determination, introducing the Effective Onduscular Wave Function which is the value of particle into the space-time at instant t into spherical, cylindrical, or Cartesian cords:

1) The Graviton-Equation

Starting from the onduscular equation we will particularize graviton, tau neutrino, photon, monopole, and electron into the potential field U(r) depending on the radius. It's not important to have the exact general solution as shown, we need the structure of the general solution of the onduscular equation.

$$\frac{\mathbf{R}}{\mathbf{r}}\Delta\mathbf{f}(\mathbf{r},\mathbf{t}) = \frac{\mathbf{1}}{\mathbf{c^2}}\frac{\partial^2\mathbf{f}(\mathbf{r},\mathbf{t})}{\partial\mathbf{t^2}} \quad (1)$$

where R= is a constant distance first we search for the basis of solutions that have separated variables, thereafter we impose the initial conditions for the graviton such as to sort out the singular solution of interest:

A) [f(r,t) = 0 for r=2·R for any t] that can be replaced with g (2·R) = 0;

B) g'(0) = R for r = 0 the initial derivate.

C) and that verifies the condition that the speed of interaction is the velocity of light c.

The above 3 conditions imposed onto the Graviton Effective Wave Function f(r,t) uniquely determine the basis of separated solutions of interest. Condition C) can be expressed employing the divergence operator in general coordinates r and c·t thus obtaining the simple equation div[f(r,c·t)]=1, expanded to

$$\frac{df(r,t)}{dt} = \frac{\partial[u(t)g(r)]}{\partial t} + \frac{\partial[u(t)g(r)]}{\partial r}\cdot\frac{\partial r}{\partial t} = c \quad (2)$$

Since the graviton speed will be c regardless of the length of the radius r, it means that: $\frac{\partial r}{\partial t} = c$ for any r, we obtain that for any r and c·t:

$$g(r)\frac{\partial u(t)}{\partial t} + u(t)\frac{\partial g(r)}{\partial r}\cdot c = c \quad (3)$$

With flux conservation of second-degree order:

$$\frac{\partial^2 P(r,t)}{c^2\partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = 0 \quad (4)$$

Thus the gravitational force F_G~flux/ surface unit = P(r,t) /4πr^2:

$$\frac{Fgflux}{4\cdot\pi\cdot r^2} = \frac{(_c_2\cdot t^2 + 4\cdot R^2 - 2\cdot R\cdot r + 2_C3\cdot t)\cdot c^2 + (-4R^2 + r^2)_c_2}{8\cdot\pi\cdot r^2\cdot c^2}$$

Considering the initial value of time t = 0, the value of Earth's gravitational force is proportional to (5) $F_{g1}\sim\frac{R^2}{2\cdot\pi}\left(1-\frac{_c_2}{c^2}\right)\frac{1}{r^2} - \frac{R}{4\cdot\pi}\frac{1}{r} + \frac{_c_2}{8\pi c^2}$

And with boundary condition P(2R,0) = 0 we have: $_c_2 = \frac{4\cdot c^2}{7}$

Thus, the unit quantic force of one proton that absorbed ν neutrino equation (6):

$$F_{g1}\sim\frac{3R^2}{14\cdot\pi}\cdot\frac{1}{r^2} - \frac{R}{4\cdot\pi}\cdot\frac{1}{r} + \frac{1}{14\cdot\pi}$$

Thus, the quantic gravitational forces in Newtons between 2 nucleons that change ν neutrino (gravitons) are (7) $F_{g1} = \frac{4.09\cdot10^{-64}}{r^2} - 1.34\cdot\frac{10^{-86}}{r} + 1.07\cdot10^{-109}\sim\frac{4.09\cdot10^{-64}}{r^2}$ thus

$F_{g1}\sim\frac{1.85\cdot10^{-64}}{r^2}\div\frac{4.09\cdot10^{-64}}{r^2}$

2) The Tau Neutrino-Equation

Starting from the onduscular equation particularized for Tau Neutrino we have:

$$\frac{R}{r}\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f(r,t)}{\partial t^2} \quad (1)$$

where R= is a constant distance first we search for the basis of solutions that have separated variables, thereafter we impose the initial conditions for the tau neutrino such as to sort out the singular solution of interest:

A) [f(r,t) = 0 for r=R for any t] that can be replaced with g (R) = 0;

B) g'(0) = R for r = 0 the initial derivate;

C) and that verifies the condition that the speed of interaction is the velocity of light c.

The above 3 conditions imposed onto the Tau Neutrino Effective Wave Function f(r,t) uniquely determine the basis of separated solutions of interest. Condition C) can be expressed employing the divergence operator in general coordinates r and c·t thus obtaining the simple equation div[f(r,c·t)]=1, expanded to:

$$\frac{df(r,t)}{dt} = \frac{\partial[u(t)g(r)]}{\partial t} + \frac{\partial[u(t)g(r)]}{\partial r}\cdot\frac{\partial r}{\partial t} = c \quad (2)$$

Since the graviton speed will be c regardless of the length of the radius r, which means that

$\frac{\partial r}{\partial t} = c$ for any r, we obtain that for any r and c·t:

$$g(r)\frac{\partial u(t)}{\partial ct} + u(t)\frac{\partial g(r)}{\partial r} = 1 \quad (3)$$

For time flow Flux Tau neutrino, we have the second order (wave equation)

$$\frac{\partial^2 P(r,t)}{c^2\partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = 0 \quad (4)$$

With initial condition P(R,t)=0 and diff(P(0,0),r)=-R and spherical cords symmetry;

Time flow ~ flux/ surface unit P(r,t) /4πr^2: (5)

$$Tflow = \frac{(-R^3\cdot r + (c^2\cdot t^2 + _C3\cdot t + _C2)R^2 - (c^2\cdot t^2 + r^2)_C2)}{4\cdot\pi\cdot r^2\cdot R^2}$$

And for t_0 = 0 the flux is proportional with:

$$TF\sim\frac{(-R^3\cdot r + (r^2 + _C2)R^2 - r^2_C2)}{4\cdot\pi\cdot r^2\cdot R^2}$$

The perception of time on Mars is slower than on Earth and considering the initial value of time t = 0 and the series development:

$$TF\sim\frac{C2}{4\cdot\pi}\cdot\frac{1}{r^2} - \frac{R}{4\cdot\pi}\cdot\frac{1}{r} + \frac{R^2 - C2}{4\pi R^2}$$

Thus, on each second on Mars, time loses ~10^{-9} sec, and time is slow on a Martian per day about 0.1336-0.0824 ns per second depending on the distance to the Sun (249 milKm or 206 milKm). The time shift is detectable with an atomic watch C2 = -1.1711403·10^{25} of dim m^2[L^2]. The flux increment is.

$$\Delta TF\sim -9.156722878\cdot\frac{10^{23}}{r^2} + 2.83852841\cdot\frac{10^{21}}{r} - 0.07957747152$$

Thus 7.52381·10^9 of Time Quanta's each one equal to 1.765·10^{-19} sec according to FluxTau.mw

We have $T_{Earth} = T_{Mars} + \Delta t$ where Δt equal to ~ ΔT = 7.52381·10^9 · 1.765·10^{-19} =1.328·10^{-9} sec.

Thus per day, the time shift is daily 0.114 - 0.0784ms between Mars and Earth relative to the distance from the Sun.

3) The Muon Neutrino-Equation

Starting from the onduscular equation particularized for Muon Neutrino we have:

$$\frac{R_s}{\mathbf{r}}\Delta\mathbf{f}(\mathbf{r},\mathbf{t}) = \frac{\mathbf{1}}{\mathbf{c^2}}\frac{\partial^2\mathbf{f}(\mathbf{r},\mathbf{t})}{\partial\mathbf{t^2}} \quad (1)$$

where Rs= is a constant distance of about ~$7.0613\cdot10^{25}$ m first we search for the basis of solutions that have separated variables, thereafter we impose the initial conditions for the muon neutrino such as to sort out the singular solution of interest:

A) [f(r,t) = 0 for r=R_s for any t] that can be replaced with g (R_s) = 0;

B) g'(0) = R_s for r = 0 the initial derivate;

C) and that verifies the condition that the speed of interaction is the velocity of light c.

The above 3 conditions imposed onto the Muon Neutrino Effective Wave Function f(r,t) uniquely determine the basis of separated solutions of interest. Condition C) can be expressed employing the divergence operator in general coordinates r and c·t thus obtaining the simple equation div[f(r,c·t)]=1, expanded to

$$\frac{df(r,t)}{dt} = \frac{\partial[u(t)g(r)]}{\partial t} + \frac{\partial[u(t)g(r)]}{\partial r}\cdot\frac{\partial r}{\partial t} = c \quad (2)$$

Since the graviton speed will be c regardless of the length of the radius r, it means that

$\frac{dr}{dt} = c$ for any r, we obtain that for any r and c·t:

$$g(r)\frac{\partial u(t)}{\partial t} + u(t)\frac{\partial g(r)}{\partial r}\cdot c = c \quad (3)$$

For the muon neutrino, the flux equation for μ neutrino considering and spherical cords symmetry & the speed of graviton equal to light speed is for (electronic neutrino) Flux second-order equation wave equation is: $\frac{\partial^2 P(r,t)}{c^2\partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = 0$ (4)

With initial condition P(2R_s,t)=0 si diff(P(0,0),r)=-R_s =-$7.061266475\cdot10^{25}$ m (7.463764553billion lightyers);

Thus the solution of Λ Space flux/ surface unit Λflux = P(r,t) /4πr^2 equation:

$$\frac{P(r,t)}{4\cdot\pi\cdot r^2} = \frac{(_c_2\cdot t^2 + 4\cdot R_s^2 - 2\cdot R_s\cdot r + 2_C3\cdot t)\cdot c^2 + (-4\cdot R_s^2 + r^2)_c_2}{8\cdot\pi\cdot r^2\cdot c^2}$$

And with initial condition t=0 (24):

$$\frac{P(r,t)}{4\cdot\pi\cdot r^2} = \frac{(4\cdot R_s^2 - 2\cdot R_s\cdot r)\cdot c^2 + (-4\cdot R_s^2 + r^2)_c_2}{8\cdot\pi\cdot r^2\cdot c^2}$$

$\int_0^{2R_s}\frac{P(r,t)}{4\cdot\pi\cdot r^2}dr = \int_0^{2R_s}(\frac{(4\cdot R_s^2-2\cdot R_s\cdot r)\cdot c^2+(-4\cdot R_s^2+r^2)_c_2}{8\cdot\pi\cdot r^2\cdot c^2})dr \sim 5.728\cdot R_s/\dim(R_s)$ the total space growth thus: $_c_2 = -9.711452983c^2$ Thus, we have equation (5):

$$\Lambda = \frac{4\cdot R_s^2 + 38.84581193\cdot R_s^2}{8\cdot\pi}\cdot\frac{1}{r^2} - \frac{R_s}{4\cdot\pi}\cdot\frac{1}{r} + \frac{9.711452983}{8\cdot\pi}$$

4) The Photon-Equation

A specific solution of the onduscular equation for the photon where R is a constant distance is

$$\frac{R}{r}\Delta H(r,\theta,z,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,z,t)}{\partial t^2}\ . \quad (5)$$

Considering that the photon is invariant of θ (z axis symmetry) in cylindrical coordinates with z axis symmetry, we are interested in searching for the separable solutions f(r,z,t) = $H_3(r)$ $H_4(z)$ u(t) with z axial symmetry z axis, which satisfy the following four singular conditions for the photon

A) [f(r,z,t) = 0 for r=R for any t and z] that can be replaced with $H_3(R) = 0$;

B) $H_3'(1/R) = R$ for r = 1/R the initial derivate;

C) and that verifies the condition that the speed of a photon is the velocity of light c.

Since$H_4(z) = -(\sin(kz) + \cos(kz))$, the wave pulsation is $k = \nu = \frac{E}{h}$, where E is the photon energy and h is the Plank constant.

Condition C) refers to the velocity of the photon which is c, the speed of light, irrespective of r and c·t.

Considering that H(r,z,θ,t) is, in fact, f(r,z,t) we have

$$\frac{df(r,z,t)}{dt} = \frac{\partial[u(t)H3(r)H4(z)]}{\partial t} + \frac{\partial[u(t)H3(r)H4(z)]}{\partial r}\cdot\frac{dr}{dt} + \frac{\partial[u(t)H3(r)H4(z)]}{\partial z}\cdot\frac{\partial z}{\partial t} = c \quad (6)$$

By dividing with c, we obtain

$$\frac{\partial[u(t)H3(r)H4(z)]}{\partial ct} + \frac{\partial[u(t)H3(r)H4(z)]}{\partial r} + \frac{\partial[u(t)H3(r)H4(z)]}{\partial z}\cdot\frac{\partial z}{\partial t}\frac{\partial t}{\partial r} = 1 \quad (7)$$

And $$\frac{\partial[u(t)H3(r)H4(z)]}{\partial ct} + \frac{\partial[u(t)H3(r)H4(z)]}{\partial r} + \frac{\partial[u(t)H3(r)H4(z)]}{\partial z}\cdot\frac{\partial z}{\partial r} = 1 \quad (8)$$

But $\frac{\partial z}{\partial r} = 0$, therefore,

$$\frac{\partial[u(t)H3(r)H4(z)]}{\partial ct} + \frac{\partial[u(t)H3(r)H4(z)]}{\partial r} = 1, \quad (9)$$

Thus $$H4(z)H3(r)\frac{\partial u(t)}{\partial ct} + u(t)H4(z)\frac{\partial[H3(r)]}{\partial r} = 1, \quad (10)$$

where the constant distance R is about $3.9228\cdot10^{22}$ m, which is slightly below 4.1465 million light-years, probably the same value as for the equation's cases for the graviton, the tau neutrino, the monopole, and the electron. Only the R_s constant from quantic space is around 16.146 billion light-years and differs from the constant R for the electronic neutrino. A simple observation of the star sky shows that there are no galaxies larger in diameter than 4.1465·4 = 16.588 million light-years, simply because after that distance the rejection of the star occurs, the circumstance that explains why the universe is accelerated expanding. The photon particles in the flux equation are in the cylindrical cords symmetry: $\frac{\partial P(r,z,t)}{c^2\partial t^2} - \frac{\partial P(r,z,t)}{\partial r^2} = 0$ (25) Where P(r,z,t)= H(r)H_1(z)U(t) Thus equation (25) dividing by H1(z) become independent of H(z) so $\frac{\partial H(r)U(t)}{c^2\partial t^2} - \frac{\partial H(r)U(t)}{\partial r^2} = 0$ simplifying H_1(z) where H(r) obeys the equation with initial condition H(1/R) = R for r = 1/R the initial value at the limit and H(R)=0.

5) The Monopole-Equation

A specific solution of the onduscular equation for the photon where R is a constant distance is

$$\frac{R}{r}\Delta H(r,\theta,z,t) = \frac{1}{c^2}\frac{\partial^2 H(r,\theta,z,t)}{\partial t^2} \quad (5bis)$$

Considering that the monopole is invariant of θ (z axis symmetry) in cylindrical coordinate with z axis symmetry, we are interested in searching for the separable solutions f(r,z,t) = $H_3(r)$ $H_4(z)$ u(t) with z axial symmetry z axis, which satisfies the following four singular conditions for the monopole; **for monopole the condition is almost the same as from the photon equation **:

A) [f(r,z,t) = 0 for r=2R for any t and z] that can be replaced with $H_3(2R) = 0$;

B) $H_3'(1/R) = R$ for r = 1/R the initial derivate;

C) And that verifies the condition that the speed of a photon is the velocity of light c

The function for z axis is $H_4(z) = -(\sin(kz) + \cos(kz))$

Considering the monopole quanta has the same energy as photon energy, h_m but with a higher value as Planck constant $E=h_m \cdot \nu$ were $h_m \sim 2.27 - 2.37 \cdot 10^{-8}$eV the constant of minimum interaction of the monopole greater than Planck constant.

Thus we have the wave pulsation of k·z of the axis of propagation thus $\frac{E}{h_m} = k$ since $H_4(z) = -(\sin(kz) + \cos(kz))$, the wave pulsation is $k = \nu = \frac{E}{h_m}$, where E is the monopole energy and h_m is a constant with the value above.

The other condition for the equation refers to the velocity of the monopole which is c the speed of light all the time therefore for the monopole we have equations.

For a monopole, particles that change the monopole's Flux equation are in cylindrical cords symmetry: $\frac{\partial P(r,z,t)}{c^2 \partial t^2} - \frac{\partial P(r,z,t)}{\partial r^2} = 0$ (25) Where P(r,z,t)= H(r)H_1(z)U(t) Thus equation (25) dividing by H1(z) become independent of H(z) so $\frac{\partial H(r)U(t)}{c^2 \partial t^2} - \frac{\partial H(r)U(t)}{\partial r^2} = 0$ simplifying $H_1(z)$ where H(r) obeys the equation with initial condition H(1/R) = R for r = 1/R the initial value at the limit and H(2R)=0 thus for u(t):

$$U(t) = C1 sin\left(\frac{c\sqrt{\frac{d^2H(r)}{dr^2}}}{\sqrt{H(r)}} \cdot t\right) + C2 cos\left(\frac{c\sqrt{\frac{d^2H(r)}{dr^2}}}{\sqrt{H(r)}} \cdot t\right)$$

6) The Electron-Equation

For the electron, we have an equation that ignores the relativistic mass effect where m, E, and V are the electron mass, energy, and velocity into the potential field U(r).

$$2m\frac{E-U(r)}{h^2} f1(r,t) + \frac{R}{r^2}\frac{\partial^2(r \cdot f1(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f1(r,t)}{\partial t^2} \quad (1)$$

With U(r)= $-e^2/r$ and total orbital angular magnetic moment L^2 we have

$$\frac{2m}{h^2}\left(E-\frac{e^2}{r}+\frac{L^2}{2mr^2}\right)f1(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f1(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f1(r,t)}{\partial t^2} \quad (2)$$

Or with $E=E_0\cdot E_R$ where Rydberg energy E_R is: $E_R=\frac{me^4}{32\pi^2\varepsilon^2h^2}$ and $L^2=l\cdot(l+1)\cdot\hbar^2$ azimuthal magnetic number l we have:

$$\frac{2m}{h^2}\left(E_0-\frac{8\cdot\pi\cdot\varepsilon}{r}+\frac{l\cdot(l+1)}{r^2}\right)f1(r,t)+\frac{R}{r^2}\frac{\partial^2(r\cdot f1(r,t))}{\partial r^2}=\frac{1}{V^2}\frac{\partial^2 f1(r,t)}{\partial t^2} \quad (3)$$

This equation is approximately the Schrodinger equation for the radius small. That is for ~ Bohr radius (rb) there is a linear transformation that for a certain R (a simple combination of constants equal to $R = 3.9228\cdot10^{22}$ m) the equation (1) and r<<R there is a linear transformation that for the change of variable $r = r_{Bohr}\,\rho$ and $h(r)=g(\rho)\frac{2R-\rho\cdot r_B}{2R}$ That for the interval of r between [0,2R] and ρ > 0 for r and ρ small the function f(r) and h(ρ) are equal and the equation equivalent is in the file OndSLP.mw and the change of variable change also the sign of member depending azimuthal quantum number because the minus into variable change (ρ·rb);

$$\frac{1}{\rho}\frac{d^2}{d\rho^2}\left(\rho h(\rho)\right)+\frac{2m}{h^2}\left(E-\frac{e^2}{\rho}\right)h(\rho) \quad (4)$$

The Schrodinger without angular magnetic dependence equation (8) is also equation (19.8) in Feynman's article**. Thus, the Quantum Torsion Field Theory includes all the classical Schrodinger equations and the calculus of the orbital of the atoms with angular dependence which is for the general solution**. Thus equation (1) is a Quantum Field Equation of electric classical Schrodinger equation, magnetic and light (cylindrical cords), and gravitational (spherical cords) quantic field for the microscopic world.

Now starting from (1) we have:

$$\frac{2m}{h^2}(E-\frac{e^2}{r}+\frac{L^2}{2mr^2})u(t)g(r)+\frac{u(t)R}{r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=\frac{g(r)}{V^2}\frac{\partial^2u(t)}{\partial t^2}\ \text{Dividing by u(t)·g(r)} \quad (5)$$

$$\frac{2m}{h^2}(E-\frac{e^2}{r}+\frac{L^2}{2mr^2})+\frac{R}{g(r)r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=\frac{1}{u(t)\cdot V^2}\frac{\partial^2u(t)}{\partial t^2} \quad (6)$$

Equation with separate variables:

$$\frac{1}{u(t)\cdot V^2}\frac{\partial^2u(t)}{\partial t^2}=k^2\ \text{With solution: } u(t)=\sinh(V\cdot k\cdot t) \quad (7)$$

$$\frac{2m}{h^2}(E-\frac{e^2}{r}+\frac{L^2}{2mr^2})+\frac{R}{g(r)r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=k^2 \quad (8)$$

$$\frac{2m}{h^2}[(E-\frac{e^2}{r}+\frac{L^2}{2mr^2})-\frac{h^2}{2m}k^2]\cdot g(r)+\frac{R}{r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2}=0 \quad (9)$$

Considering electron into the potential field $U(r)=-\frac{e^2}{r}$ and $E_{Rydberg}$ energy $E_{Rydberg}$

$E_R = \frac{me^4}{32\pi^2\varepsilon_0^2 h^2}$ = 13,6eV or 2.179·10^{-18} J and $r_B = \frac{4\pi\varepsilon_0 h^2}{me^2}$ radius r_{Bohr}=0,529Angstrom

$$\frac{2m}{h^2}\left[\left(E - \frac{e^2}{r} + \frac{L^2}{2mr^2}\right) - \frac{h^2}{2m}k^2\right]\cdot g(r) + \frac{R}{r^2}\frac{\partial^2(r\cdot g(r))}{\partial r^2} = 0 \qquad (10)$$

And we have the Feynman formula for the electron of a hydrogen atom (8) we change notation:

$$\frac{1}{\rho}\frac{d^2}{d\rho^2}\left(\rho g(\rho)\right) + \frac{2m}{h^2}\left(E - \frac{e^2}{\rho}\right)g(\rho) \qquad (11)$$

1. Basic equations (Maple notations: h = ħ)

The Onuscular Equation (eq6 from Maple):

$$\frac{2m}{\hbar^2}\left(E - \frac{ec^2}{\rho} + \frac{L^2}{2m\rho^2} - \frac{\hbar^2 k}{2m}\right)\frac{\rho^2 f(\rho)}{\boldsymbol{R}} + \frac{d^2(\rho f(\rho))}{d\rho^2} = 0$$

Schrödinger equation (eq1 in Maple, section two):

$$\frac{2m}{\hbar^2}\left(E - \frac{ec^2}{\rho}\right)\rho f(\rho) + \frac{d^2(\rho f(\rho))}{d\rho^2} = 0$$

2. Variable change and scaling (rb = Bohr radius)

Substitutions shall apply:

$$\rho = r_b \cdot r, \quad f(\rho) = g(r), \quad E = \frac{m\,ec^4}{32\pi^2\varepsilon^2\hbar^2}E_0, \quad r_b = \frac{4\pi\varepsilon\hbar^2}{m\,ec^2}$$

The scaled Schrödinger equation becomes (eq2, final form):

$$\left(E_0 - \frac{8\pi\varepsilon}{r}\right)r\,g(r) + \frac{d^2(r\,g(r))}{dr^2} = 0$$

Scaled onnduscular equation (onduscular eq2):

$$\left(\frac{E_0 r^2}{R} - \frac{16\hbar^4 k\pi^2\varepsilon^2 r^2}{m^2 ec^4 R} - \frac{8\pi\varepsilon\, r}{R} + \frac{L^2}{\hbar^2 R}\right)g(r) + 2\frac{dg}{dr} + r\frac{d^2 g}{dr^2} = 0$$

3. Changing variables with the limit factor (border condition)

The zero transformation to r = 2R is applied:

$$f(\rho) = g(r)\cdot\frac{2R - r\cdot r_b}{2R}$$

After substitution and simplification (eq2 → eq3 in Maple), the equation becomes:

$$r\frac{d^2 g}{dr^2} + 2\frac{dg}{dr} + \mathcal{T}(r)\cdot g(r) = 0$$

where the term free expands in Taylor series around :$\mathcal{T}(r)r = 0$

$$\mathcal{T}(r) = \underbrace{\frac{r_b(16\pi R\varepsilon - 2)}{2R}}_{\text{ord. 0}} + \underbrace{\frac{r_b(2E_0 R\, r_b - 8\pi\varepsilon\, r_b) + \frac{r_b^2(8\pi R\varepsilon - 1)}{R}}{2R}}_{\text{ord. 1}}\cdot r + \underbrace{\frac{-r_b^3 E_0 + \frac{r_b^3(2E_0 R^2 - 1)}{2R^2}}{2R}}_{\text{ord. 2}}\cdot r^2$$

4. The condition of overlap with Schrödinger

Comparing the coefficients with the scaled onuscular equation:

$$\frac{E_0 r^2}{R} - \frac{16\hbar^4 k\pi^2\varepsilon^2 r^2}{m^2 ec^4 R} - \frac{8\pi\varepsilon r}{R} + \frac{L^2}{\hbar^2 R}$$

3 equations are obtained by identifying the coefficients at each order in: r

Order	Matching equation
r^0 (constant)	$\frac{r_b(16\pi R\varepsilon - 2)}{2R} = \frac{L^2}{\hbar^2 R}$
r^1 (linear)	$\frac{r_b(2E_0 R\, r_b - 8\pi\varepsilon r_b) + \frac{r_b^2(8\pi R\varepsilon - 1)}{R}}{2R} = -\frac{8\pi\varepsilon}{R}$
r^2 (quadratic)	$\frac{1}{2}\cdot\frac{-r_b^3 E_0 + \frac{r_b^3(2E_0 R^2 - 1)}{2R^2}}{R} = \frac{E_0}{R} - \frac{16\hbar^4 k\pi^2\varepsilon^2}{m^2 ec^4 R}$

5. Analytical solution wes.mw (31)

Maple solves the system of 3 equations in and gets: R, k $\boxed{R = \frac{r_b \hbar^2 + L^2}{8\pi\varepsilon\hbar^2 r_b}}$

or equivalent (eq. 32 of Maple): $R = \frac{r_b h^2 + L^2}{8\pi\varepsilon h^2 r_b}$

where h = ħ (Maple notation).

The simplified form (Maple equation 37), for the orbital quantum number:l $\boxed{R = \frac{l(l+1)}{8\pi\varepsilon r_b}}$

This results from the identity , as follows:$L^2 = l(l+1)\hbar^2$

$R = \frac{r_b\hbar^2 + l(l+1)\hbar^2}{8\pi\varepsilon\hbar^2 r_b} = \frac{l(l+1) + 1\cdot(r_b/r_b)}{8\pi\varepsilon r_b} \approx \frac{l(l+1)}{8\pi\varepsilon r_b} \quad (l \gg 1)$

For a general solution, we search for the separated solutions $f_{(r,t)}$=g(r)u(t) of this equation, that satisfies the following four singular conditions:

A) the initial condition g(R)=0 and $g(r_B)$=0;

B) g'(0) = R for r = 0 the initial derivate;

C) and that verifies the condition that the velocity of the electron is $V = \frac{\partial f(r,t)}{\partial t}$ thus

$\frac{df(r,t)}{dt} = \frac{\partial[u(t)g(r)]}{\partial t} + \frac{\partial[u(t)g(r)]}{\partial r}\cdot\frac{\partial r}{\partial t} = V$

According to Einstein's relativity, the mass is multiplied by a factor of $\sqrt{1 - \frac{V^2}{c^2}}$ therefore we have:

$$2m\frac{E-U(r)}{h}\sqrt{1-\frac{V^2}{c^2}}f(r,t) + \frac{R}{r^2}\frac{\partial^2(r\cdot f(r,t))}{\partial r^2} = \frac{1}{V^2}\frac{\partial^2 f(r,t)}{\partial t^2} \qquad (12)$$

Thus, condition B) and C) initial value condition: g(R) = 0, D(g)(0) = R, f(rb) = 0; and the consistent condition of velocity:

$$V = u(t)\cdot\frac{\partial g(r)}{\partial r}V + g(r)\cdot\frac{\partial g(r)}{\partial t} \qquad (13)$$

The flux equation of the electron is as in Bohm's Quantum Potential of the probability density:

The flux conservation equation for electron the equation of continuity of probability density

"The Theory of Space, Time and Gravitation" of Vladimir Aleksandrovici Fock (66.09) page 248 is: $\frac{\partial P(r,t)}{\partial t} + \frac{\partial(P(r,t)\cdot V)}{\partial r} = 0$ (14)

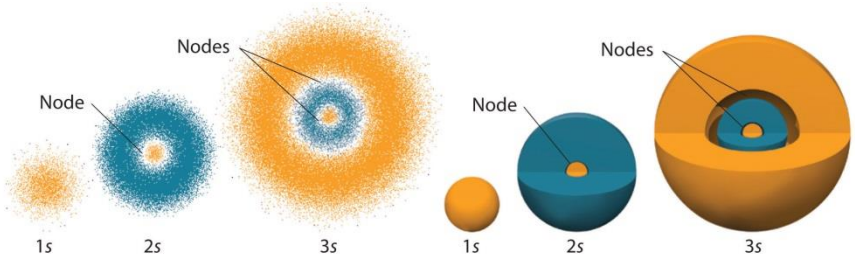
A system with an equation of second order: $\frac{\partial^2 P(r,t)}{\partial t^2} + \frac{\partial^2(P(r,t)\cdot V^2)}{\partial r^2} = \frac{V^2}{R}P(r,t)$ (15)

And considering V = constant we have the system with a nonhomogenous wave equation & the equation continuity: $\frac{\partial^2 P(r,t)}{V^2\partial t^2} - \frac{\partial^2 P(r,t)}{\partial r^2} = \frac{P(r,t)}{R}$ & $\frac{\partial P(r,t)}{V\partial t} + \frac{\partial P(r,t)}{\partial r} = 0$ (16)

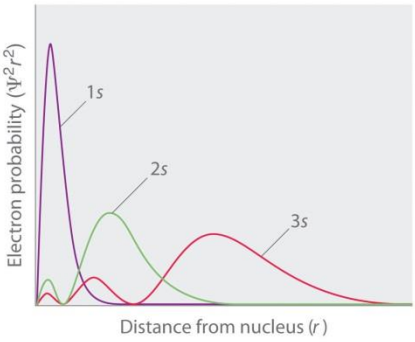
With initial condition P(R,0) = 0 and diff(P(0,0), r) = - R

Thus, the solution is (4):

$$P_e(r,t) = \frac{\sqrt{2}\cdot R^{3/2}\cdot\left(e^{\frac{(V\cdot t+R-r)\cdot\sqrt{2}}{\sqrt{R}}} - 1\right)\cdot e^{-\frac{\sqrt{2}\cdot(V\cdot t-r)}{2\cdot\sqrt{R}}}}{e^{\sqrt{2\cdot R}+1}}$$



(a) Electron probability (b) Contour probability



(c) Radial probability

24. Feb. 2026